

LIFE PROCESSES

- Living organisms: have cellular organisation, metabolism, consciousness etc.

NOTE: visible movement is not a defining feature of living organisms ∴ plants & sleeping animals are also living

- Virus is a **CONNECTING LINK** between living & non-living ∴ it is in between outside host cell

WHAT ARE LIFE PROCESSES?

- Body of organisms needs continuous repair & maintenance which requires energy.
- ALL these processes involved in this maintenance job are called life processes e.g., nutrition, respiration, excretion etc.

■ ORGANISMS

- 1 Unicellular: their entire body surface is in contact with environment ∴ no specialised structures for digestion, respiration & excretion

- 2 Multicellular: as all cells can't be in contact with environment ∴ specialised structures needed for different functions

NUTRITION

- Process of taking food & utilising it



To make energy/ATP which is needed to maintain state of order in our body

NUTRITION

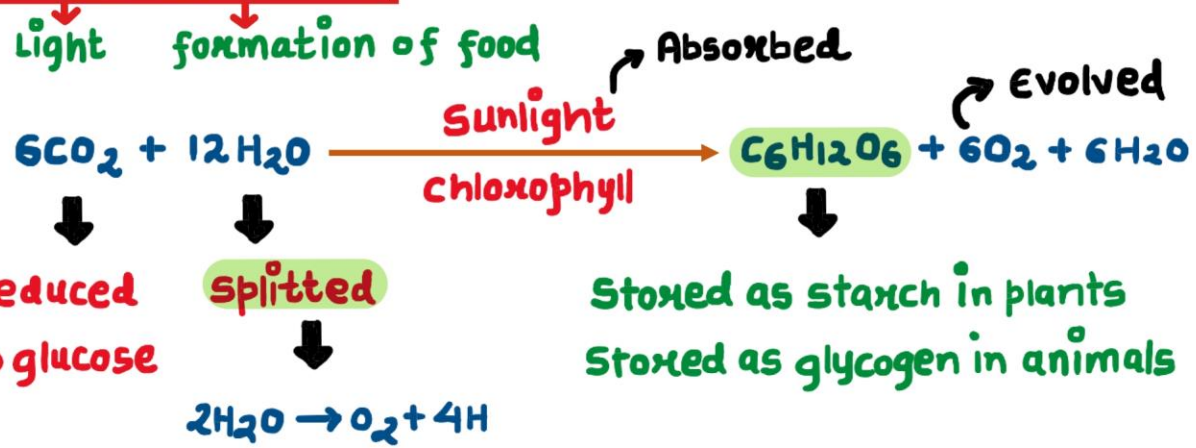
AUTOTROPHIC

- can make their own food from simple inorganic molecules like CO_2 & H_2O
- e.g., Plants & Some bacteria like cyanobacteria

HETEROTROPHIC

- Depends directly or indirectly on autotroph
- ↓
- e.g., Animals & Fungi

PHOTOSYNTHESIS

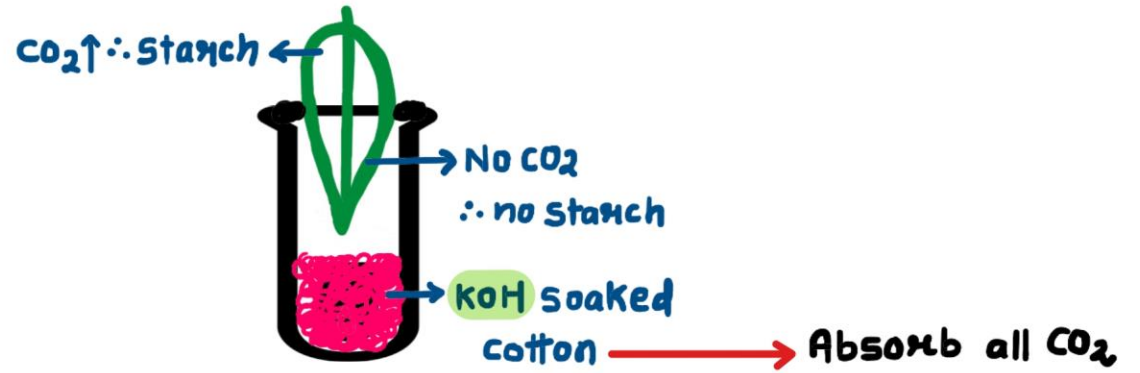


NOTE: Desert plants take CO₂ at night; make an intermediate product (organic acid); later this product is used to make sugars in presence of light

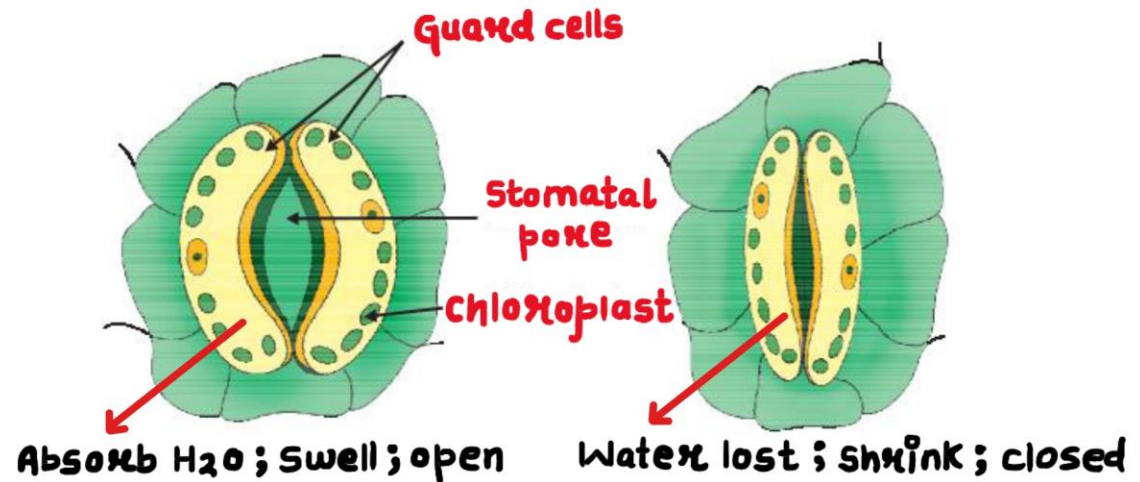
VARIEGATED LEAF EXPERIMENT



MOLL'S HALF LEAF EXPERIMENT



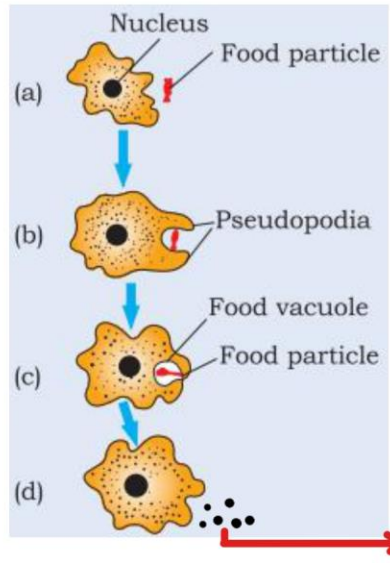
• Max. gas exchange in plants occurs through stomata



• Plants absorb H₂O & minerals (like N, P, Fe, Mg) from soil via roots

Absorbed as Nitrate or Nitrite

NUTRITION IN AMOEBA



Heterotrophic Nutrition

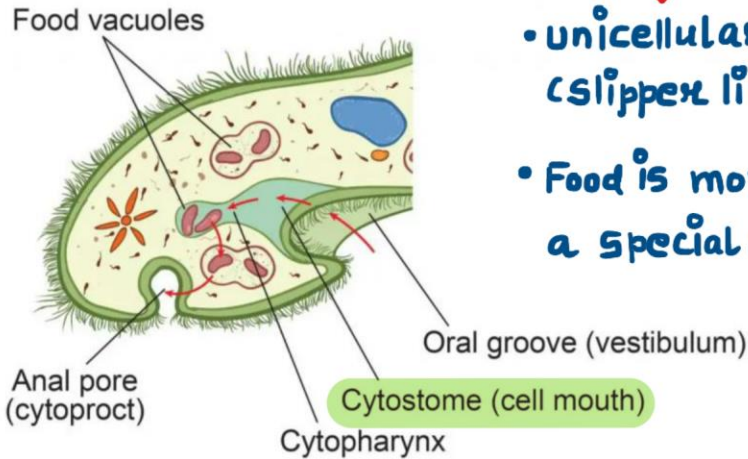
Fungi: Saprophytic Nutrition: Grow on dead and decaying matter: Bread Mould, Mushroom, Yeast

Animals: Holozoic Nutrition (first ingest, then digest)

Parasites: Cuscuta (amarbel), Ticks, Lice, Leech, Tapeworm

Different organisms have different modes!

NUTRITION IN PARAMECIUM

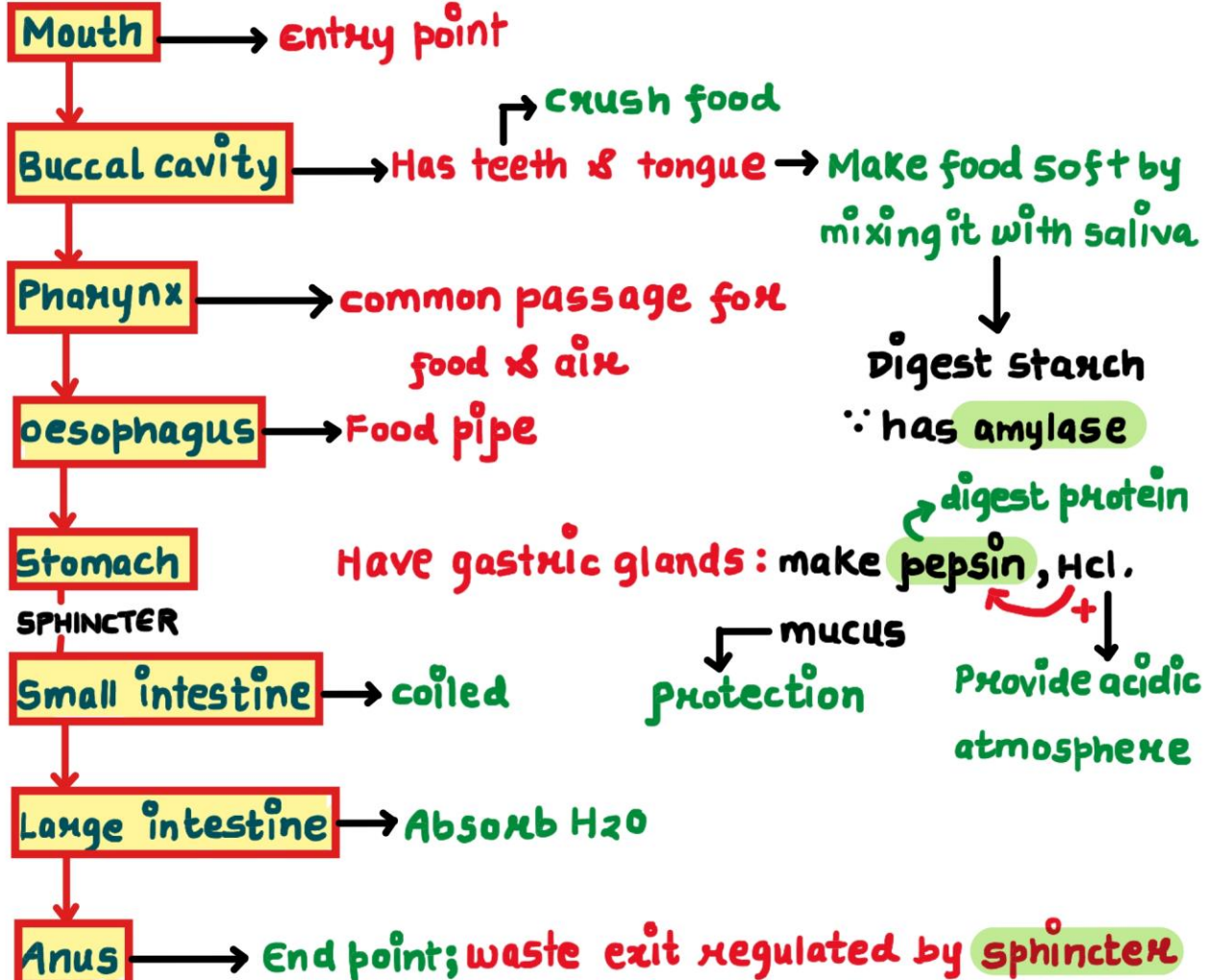


- unicellular ; have definite (slipper like) shape
- Food is moved by cilia & taken by a special spot called **cytostome**

NUTRITION IN HUMANS

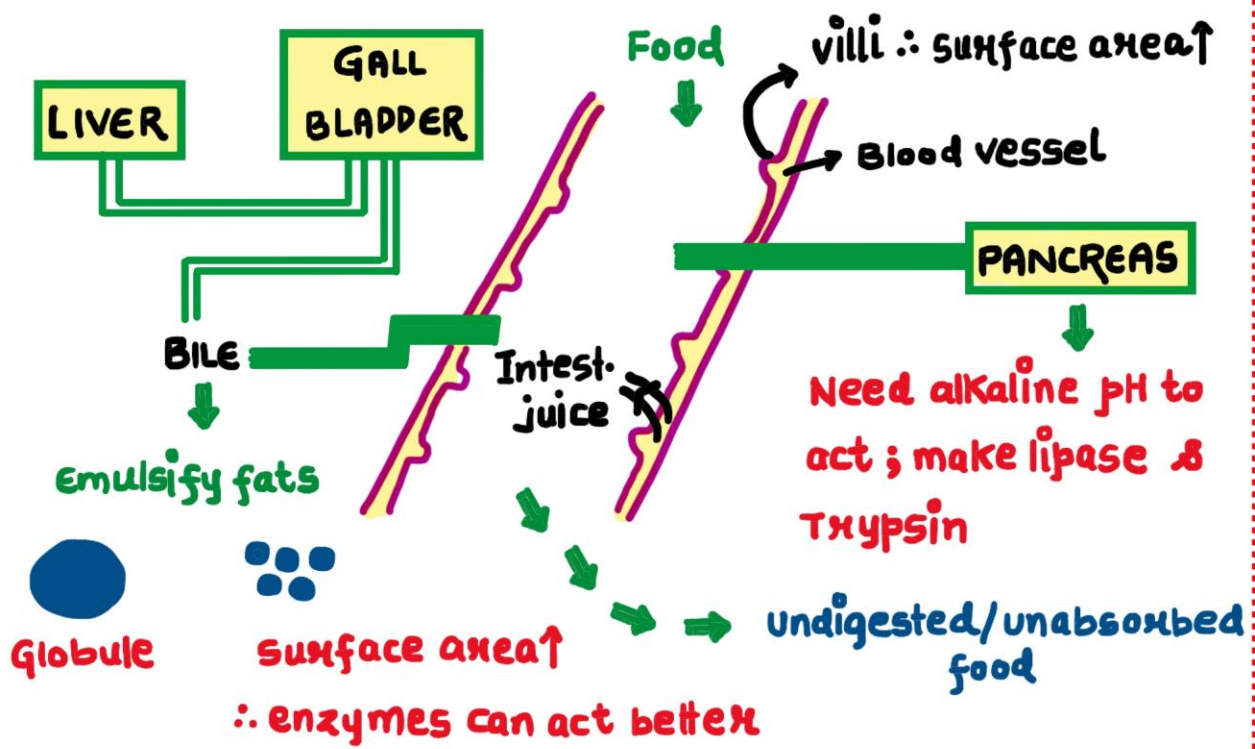
- Alimentary canal is a tube from mouth to anus.

PATH OF FOOD

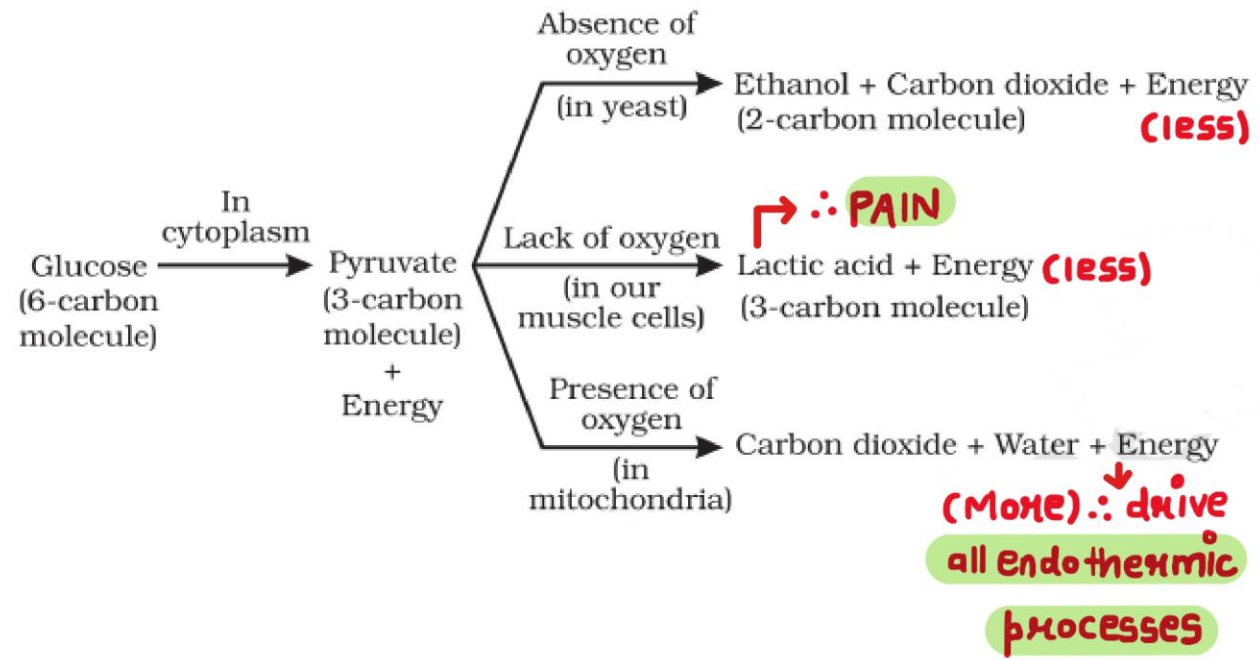
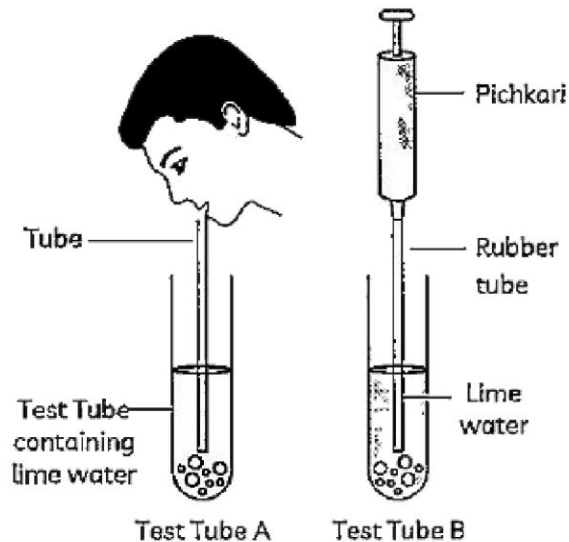


DENTAL CARIES/TOOTH DECAY

Bacteria with food stick on teeth & secrete acids that demineralise enamel ∴ Brushing is important



RESPIRATION



ATP

→ **ADP + P_i = 30.5 KJ/mol energy**
Broken and used for all body functions

∴ **O₂** is needed for respiration ∴ it needs to be inhaled
Enter in plants via stomata

Day Time	O ₂ released	CO ₂ used for photosynthesis
Night Time	CO ₂ released	No Photosynthesis

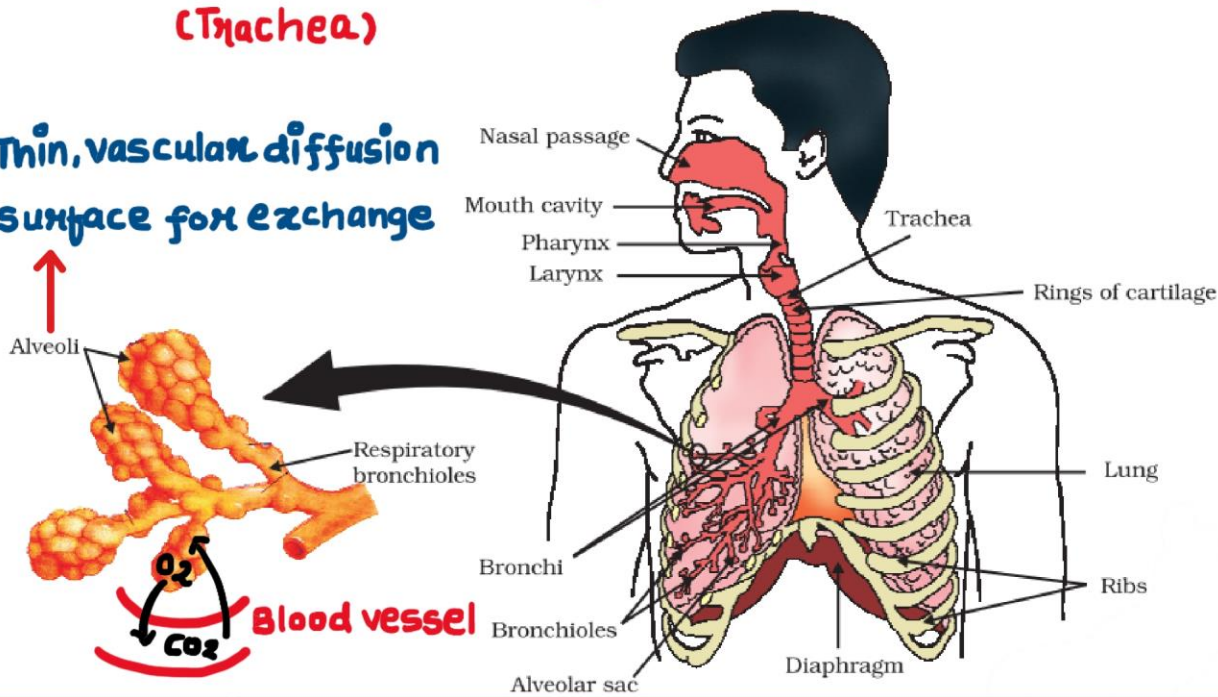
- Animals → **Terrestrial**: take O₂ from air
- **Aquatic**: take dissolved O₂ from H₂O

NOTE: Aquatic animals breathe faster than terrestrial animals

RESPIRATION IN TERRESTRIAL ANIMALS

- Nostril → Passage / Tract → Lungs
 - has alveoli
 - surface area = $80m^2$
 - Have mucus & hairs
 - Throat has cartilaginous rings (Trachea)

Thin, vascular diffusion surface for exchange



Inhalation	Diaphragm moves down; chest expands
Exhalation	Diaphragm relaxes; chest cavity go back to original position

NOTE: Haemoglobin is respiratory pigment **ent** in RBC; transport O_2 ; provide red colour to blood

TRANSPORTATION IN HUMAN BEINGS

- Human circulatory system consists of:

1 **Blood** = Plasma + cells diffused in it

Fluid medium; transports nutrients, O_2 , CO_2 & wastes

via RBC
via plasma

2 **Vessels** = Artery + veins + capillaries → Thinnest; exchange ↑

Blood towards organs; thick, elastic walls; P ↑

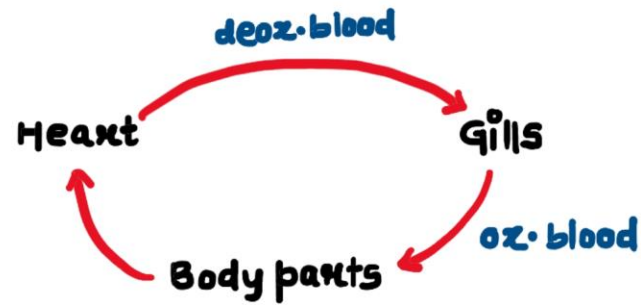
Towards heart; P ↓

3 **Heart** = 4 chambered muscular organ (size = our fist)

2 small atria & 2 large ventricles

Fishes	Amphibians, Reptiles	Birds and Mammals
2 Chambered Heart	3 Chambered Heart	4 Chambered Heart
Single circulation	Incomplete double circulation	Double circulation

Circulation in fishes:

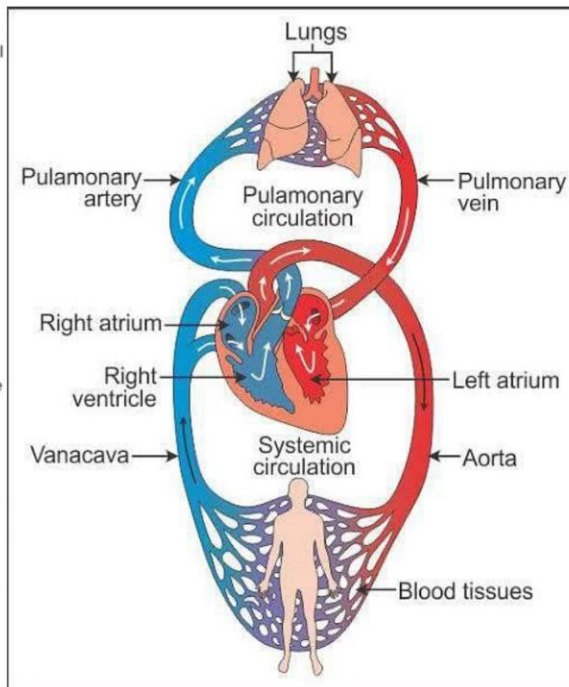
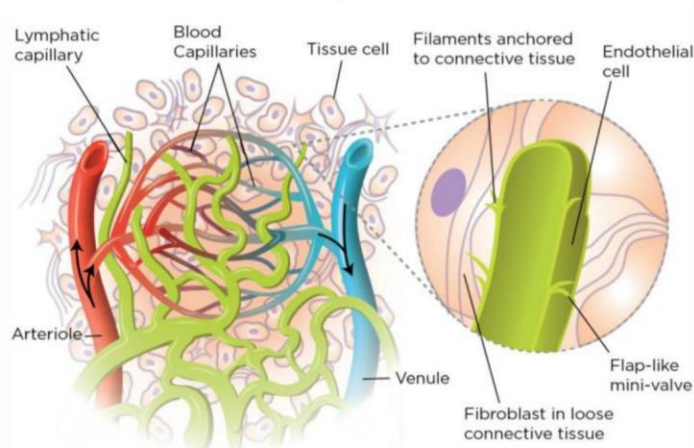


In humans: oxygenated & deox. blood flows separately

LaO RaDO

Platelets: make clot over injury & stop bleeding

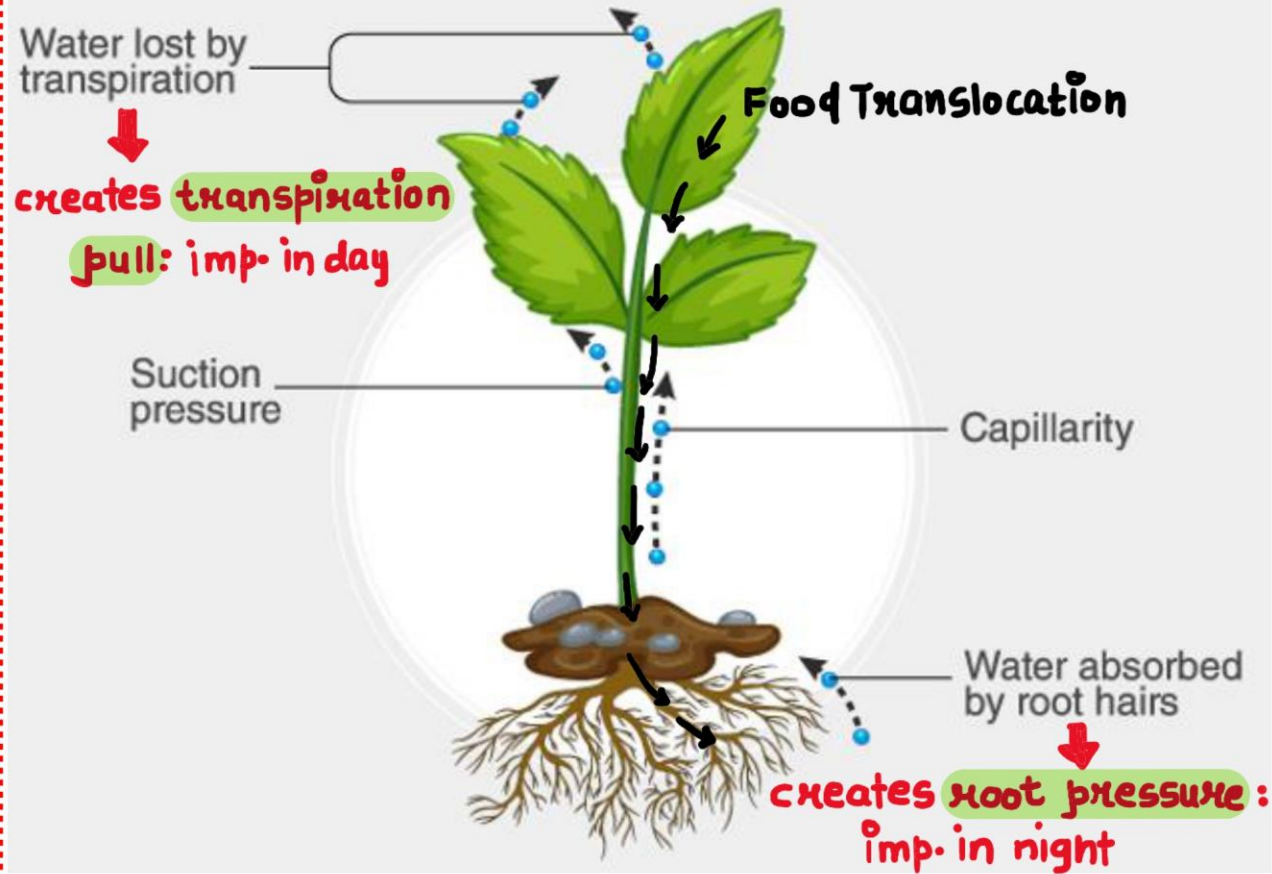
Lymph: also called Tissue fluid; contain plasma, some cells & few proteins (small)



NOTE: It also carries digested & absorbed fats from intestine

TRANSPORT IN PLANTS

• Soil is the nearest & richest source of raw materials

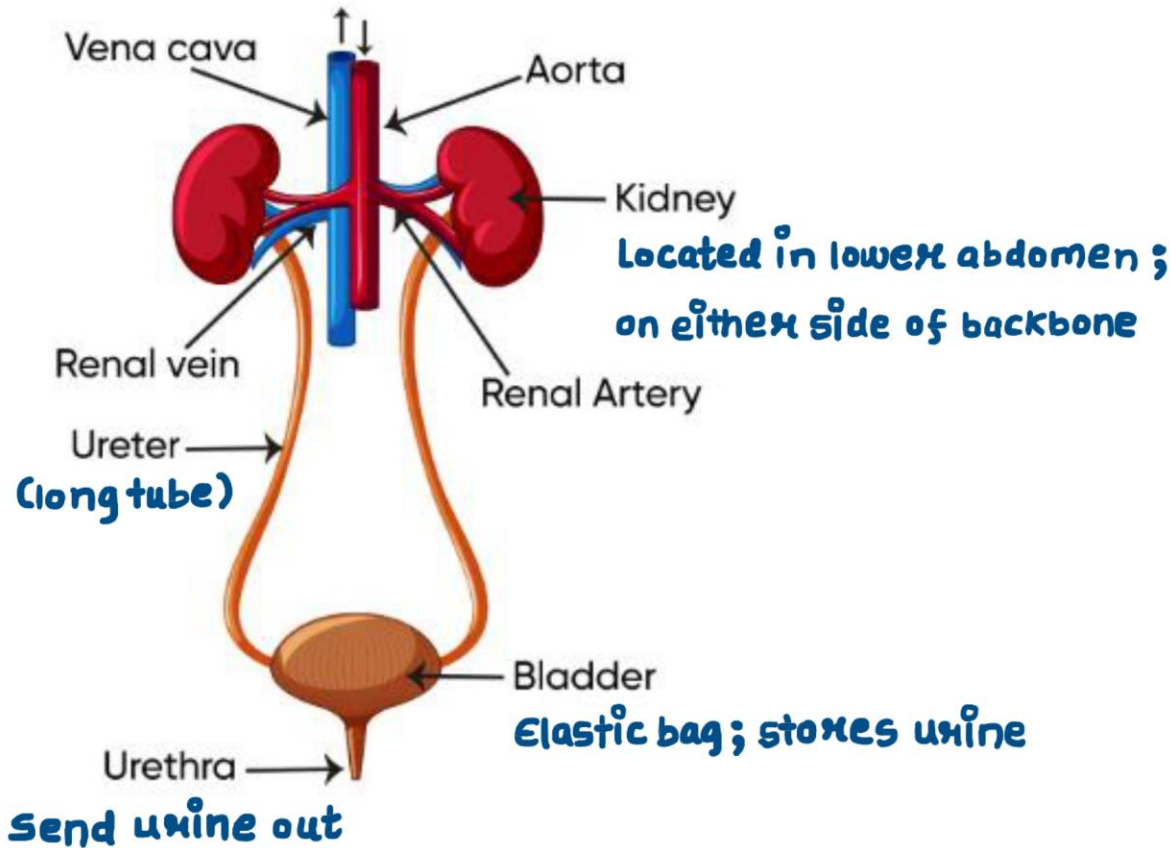


• Xylem (mainly tracheid & vessel) transport H₂O & minerals from root to stem

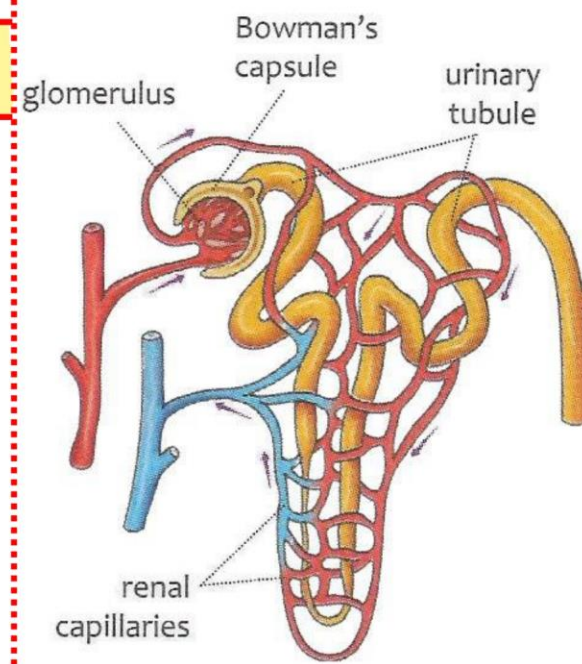
• Phloem (mainly sieve tube & companion cells) transport food, amino acids from leaf to root, stem, fruit, seed etc.

EXCRETION IN HUMAN BEINGS

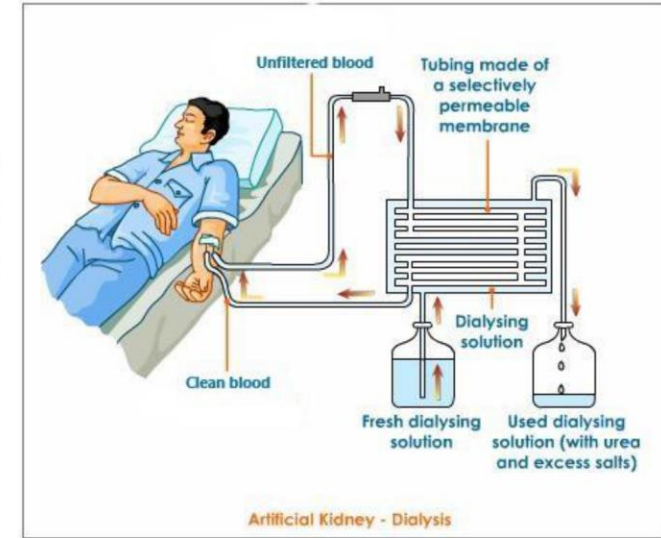
- **Excretion**: process of eliminating Nitrogenous waste from body



- **Nephron**: structural & functional unit of kidney
- **Nephron**: Bowman's capsule + Tubular part (cup shaped) (Tubes)



- Glucose, amino acid, H_2O , Salts are reabsorbed



EXCRETION IN PLANTS

Gaseous Exchange	Stomatal Pore
Excess Water	Transpiration
Many waste products are even stored in	Vacuoles and in leaves that falls off
Wastes may be stored as resin and gums in	Old xylem
Some waste products are excreted	Directly in soil

CONTROL AND COORDINATION

- In response to **environmental changes**, the organism shows various **movements**



May lead to **growth**
e.g., plants showing **phototropism**

Do not lead to **growth**
e.g., children running

- All events of our body are under **neural & hormonal (endocrine) control**
- All organs of our body **complement/coordinate** with each other for effective functioning

ANIMALS: NERVOUS SYSTEM

- Our nervous tissue has network of specialised cells called **nerve cells/ neurons**



conduct electrical impulse from one part to next part

Gustatory receptor detects **taste** & **olfactory** receptor detects **smell**

- Flow of information:**

Information received by tip of dendrite



Impulse sent to **cyton/cell body**



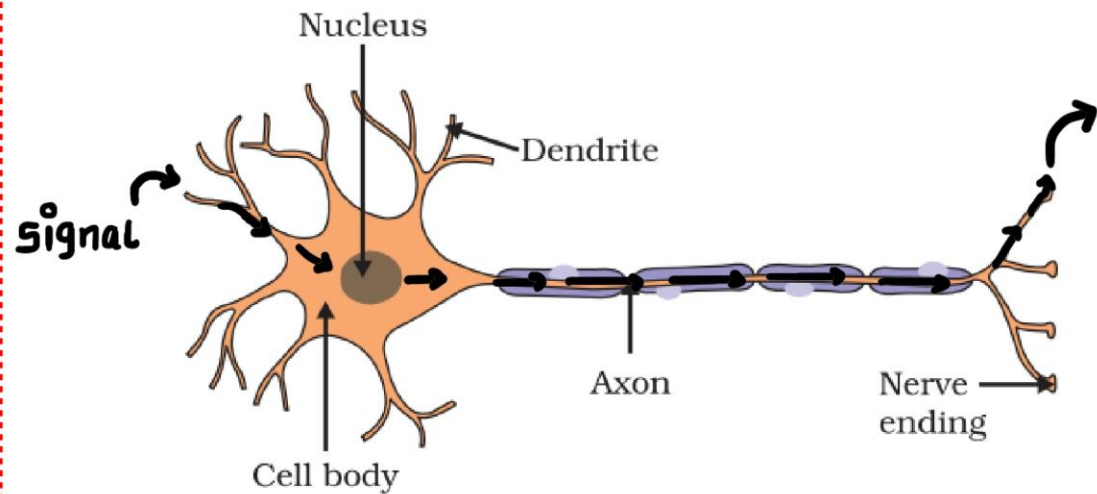
Impulse sent to **axon** & then to **nerve endings**



Neurotransmitters (NT) released



NT **cross synapse** & generate **new impulse** in next **neuron/gland/muscle**



REFLEX ACTION

- Sudden reaction in response to any stimulus without thinking

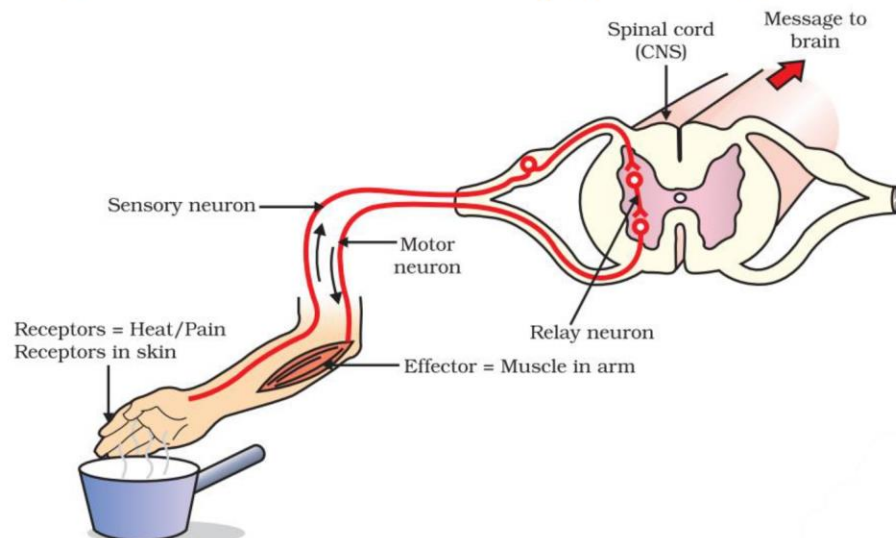
- Reflex arc has evolved as efficient way to function in absence of true thought process ∴ less time ↓

REFLEX ARC

Receptor → Sensory Neuron → Inter/Relay neuron/
Spinal cord

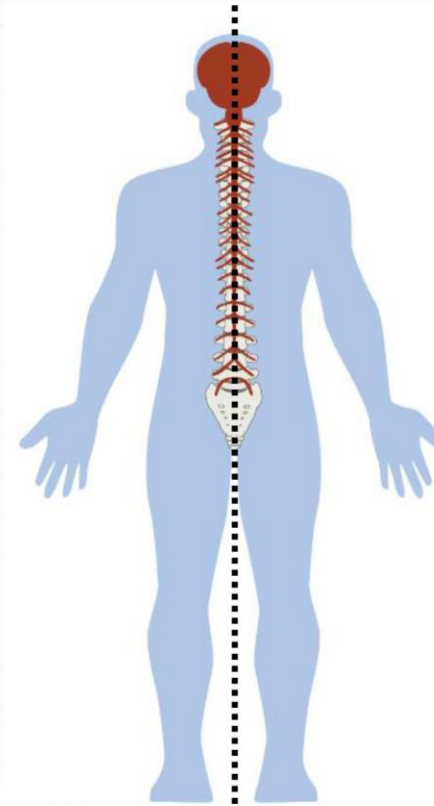
Have special proteins that can change shape
← Muscle ← Motor Neuron

e.g., salivation on seeing food; eye blink → cornea touched

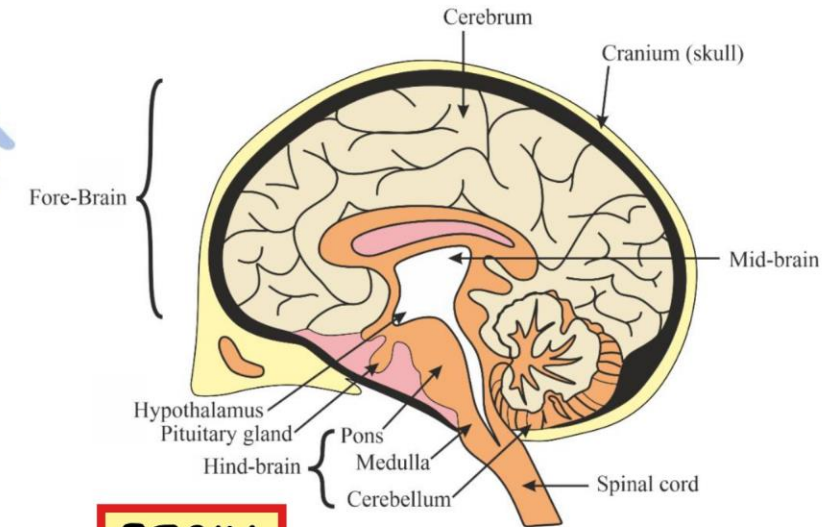


- Main thinking part;
- Hearing, smell, sight
- Vol. movem.

HUMAN BRAIN



- Main coordinating centre
- **CNS** = Brain + spinal cord (Cranium) (vertebral column)
- **PNS** = cranial + spinal nerves



BRAIN

FOREBRAIN

- cerebrum
- Thalamus
- Hypothalamus → Heat-drink

MIDBRAIN

Inv. movements along with Hindbrain

Balance

BP, salivation, vomiting ↓

HINDBRAIN

- Pons → Sleep
- cerebellum
- Medulla

COORDINATION IN PLANTS

- Plants do not have any neural & muscular system
- Plant cells change their shape by changing amount of H_2O in them $\therefore$ they **swell or shrink**
- e.g., folding of leaves in Mimosa i.e., Chhui mui plant

MOVEMENT DUE TO GROWTH



Positively phototropic
Negatively geotropic

- chemotaxis**: movement towards chemicals e.g., pollen tube

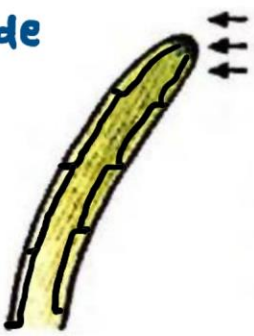
PLANT HORMONES/ PHYTOHORMONES

- Simple chemicals that move from their site of synthesis to their site of action & regulate some physiological process
- Auxin**: synthesised by growing shoot tips
- Plant appears to bend towards light; auxin diffuses to

the shady side & help cells grow on that side

2 **Gibberellin**: stem elongation $\uparrow$

3 **Cytokinin**: cell division $\uparrow$; formed at areas of great cell division like fruit & seed

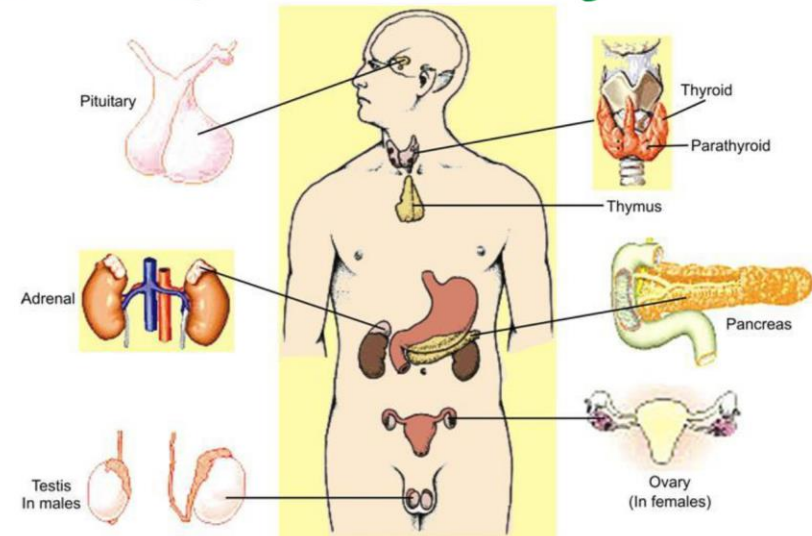


4 **Abscissic acid**: stress hormone; inhibitory; wilt leaf

5 **Ethylene**: gaseous; helps in fruit ripening

HORMONES OF ANIMALS

- Simple chemicals made by endocrine glands which travel via blood stream; secreted in minute quantity as they act specifically



Hormone	Made by	Functions	Diseases
Growth Hormone (GH)	Pituitary Gland	Stimulates growth in all organs	Dwarfism and gigantism
Insulin	Pancreas	Regulates blood sugar levels	Diabetes mellitus (sugar)
Testosterone	Testis	Development of male sex organs and regulates sexual behaviour	
Oestrogen	Ovary	Development of female sex organs; regulates sexual behaviour and menstrual cycle	

Hormone	Made by	Functions	Diseases
Adrenaline	Adrenal Gland	Fight and Flight hormone	
Releasing Hormones (RH)	Hypothalamus	Stimulate pituitary gland to release hormones	
Thyroxine	Thyroid Gland	Regulates carb, fat and protein metabolism	Goitre (swollen neck) due to Iodine deficiency

HOW DO ORGANISMS REPRODUCE?

- Reproduction is energy consuming process & it's not necessary to maintain life but is still necessary for continuity of species.
- Reproduction creates new individuals with similar body designs & with some variations
- DNA has information to make proteins in a cell; if sequence of DNA is changed, the protein formed will also be different

EVENTS IN REPRODUCTION OF A CELL

1 Copies of DNA created



2 Copies of cellular apparatus/organelles created



3 Material divided and 2 cells formed

NOTE: During DNA replication, similar copies of DNA are made which are not exactly identical

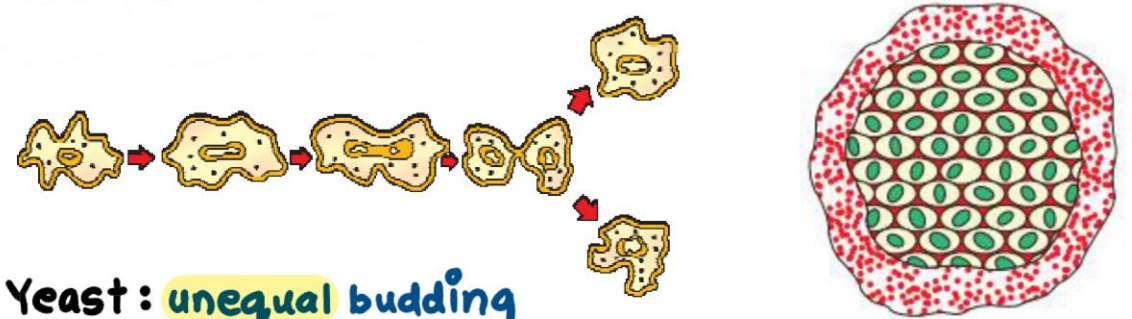
THESE VARIATIONS ARE RAW MATERIALS FOR EVOLUTION

- These variations help us adapt better & provide resistance to harsh conditions

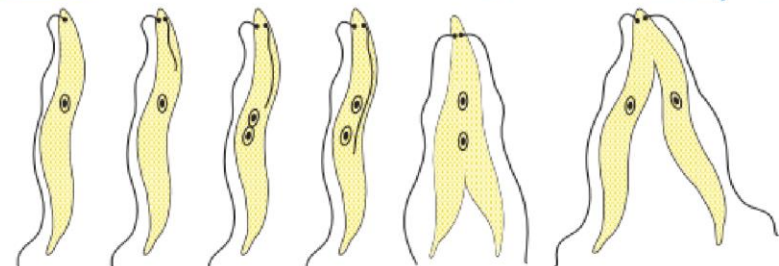
ASEXUAL REPRODUCTION (UNIPARENTAL)

- Mode of reproduction depends on body type of organism

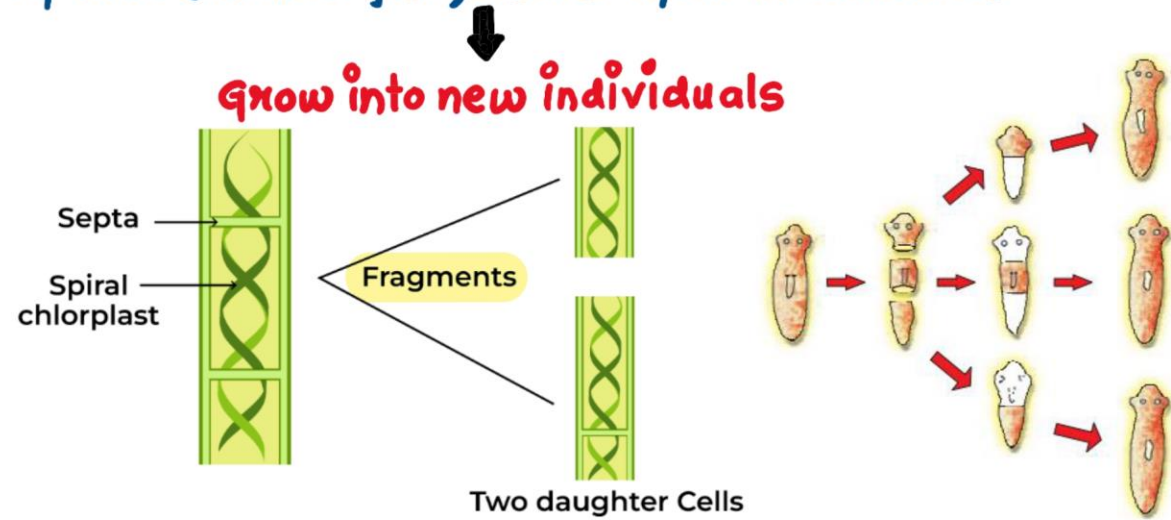
1 **FISSION**: common in unicellular organisms like bacteria & protozoa; e.g., Amoeba can split along any plane to create 2 equal halves



- Yeast: unequal budding
- Leishmania (cause 'kala azar') show long binary fission
- Plasmodium (cause 'malaria') show multiple fission

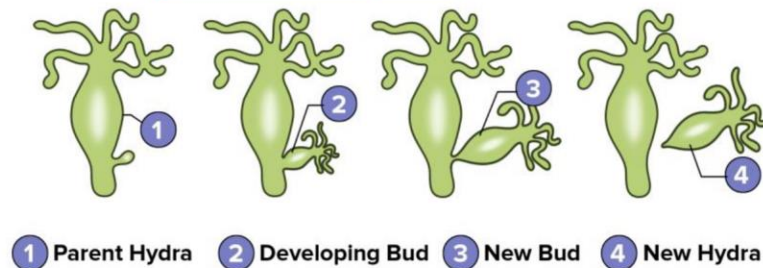


2 FRAGMENTATION: Shown by multicellular organism with simple body organisation e.g., *Spirogyra* breaks up into smaller fragments upon maturation



3 REGENERATION: cut parts of *Hydra* & *Planaria* can regenerate their complete body; specialised cells in them divide & develop

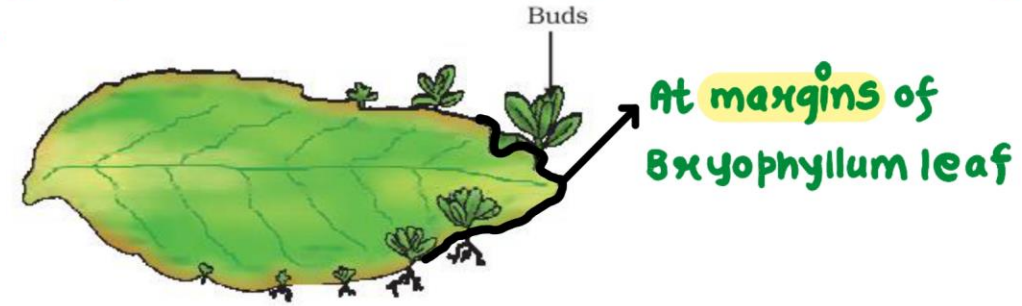
4 BUDDING: In yeast & *Hydra*



5 VEGETATIVE PROPAGATION: new plant arise

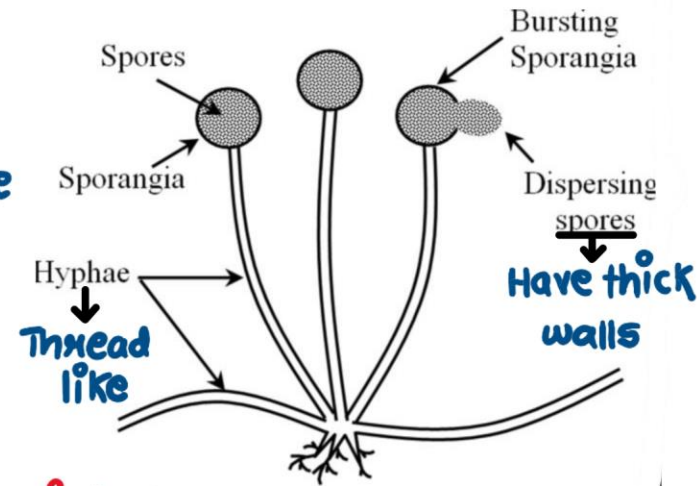
from vegetative plant part like root, stem, leaf

- Layering & grafting used to grow rose, sugarcane, grapes etc.
- Fast way to produce flowers & fruits; genetically similar



6 SPORE FORMATION:

↓ Blobs like
In simple multicellular organisms like *Rhizopus* (Bread mould fungi)

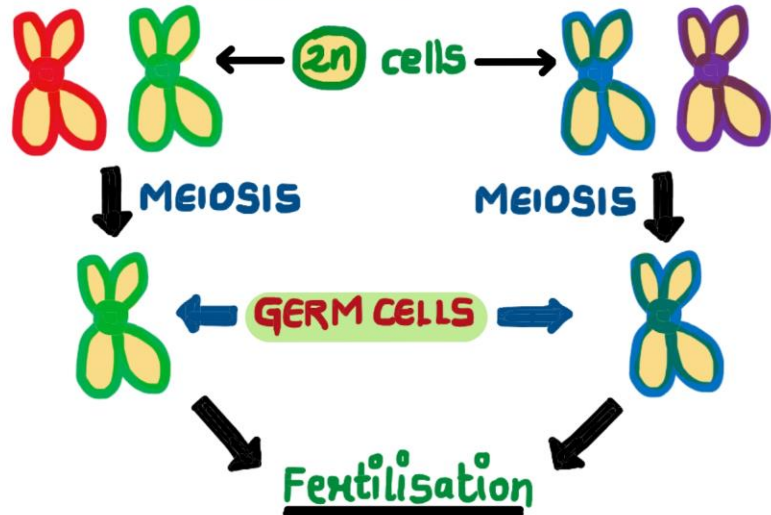


↓ grow at warm & humid/moist places

- **TISSUE CULTURE:** making many plants (similar) from a plant; disease free

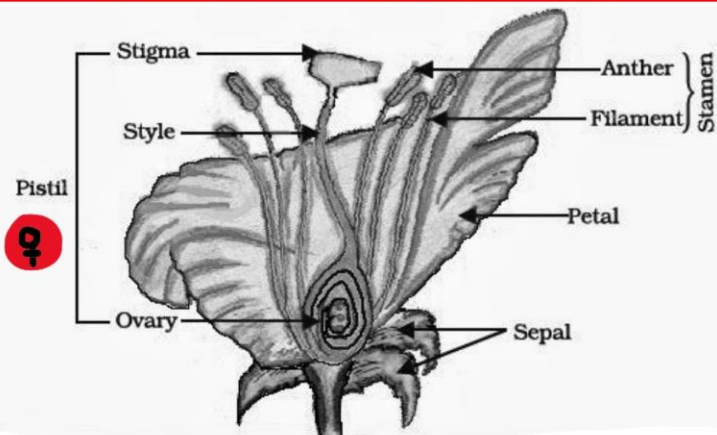
SEXUAL REPRODUCTION (BIPARENTAL)

- Slow process ; variations ↑ ; new combinations formed



- ♂-gamete: small, motile
- ♀-gamete: Large, non-motile, stores food

SEXUAL REPRODUCTION IN FLOW. PLANTS

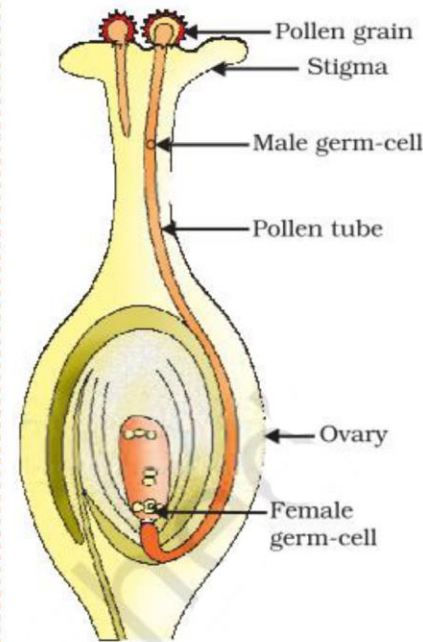


- Flower: reproductive structure
- consist: sepal, petal, stamen, pistil

Unisexual plants: Papaya, Watermelon

Bisexual: Hibiscus, Mustard

- Pollen can be transferred from same plant (self) or diff. plant (cross pollination) to female part via agents like wind, water & animals

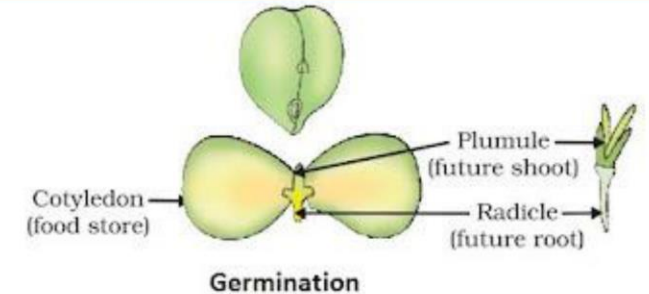


After Fertilization

Ovule → Seed

Ovary → Fruit

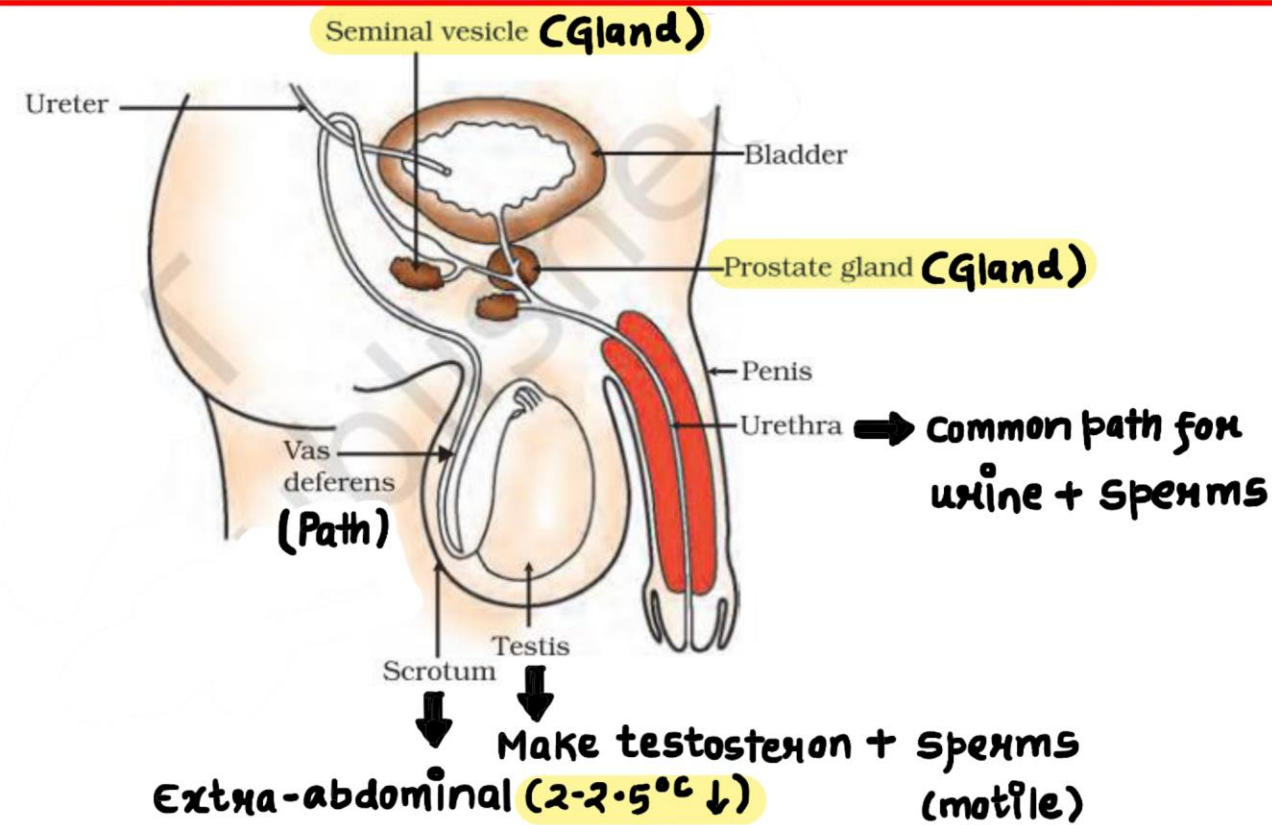
Sepals, petals, stamen, stigma and style may fall off



REPRODUCTION IN HUMAN BEINGS

- At Puberty, the rate of general body growth begins to slow down & reproductive tissue begin to mature
- Many changes are seen in ♂ & ♀ at this time

MALE REPRODUCTIVE SYSTEM



- At puberty, immature eggs starts maturing & one egg is released every month
- When egg is fertilised, it is implanted in uterus & a disc like placenta is formed
- Placenta contains villi on embryo's side & blood spaces on mother's side ∴ nutrition & exchange occur

WHAT HAPPENS TO EGG?

- If fertilised: uterus becomes thick & spongy ∴ implant.
- If not fertilised: lining is shed; menstruation (2-8 day)

REPRODUCTIVE HEALTH

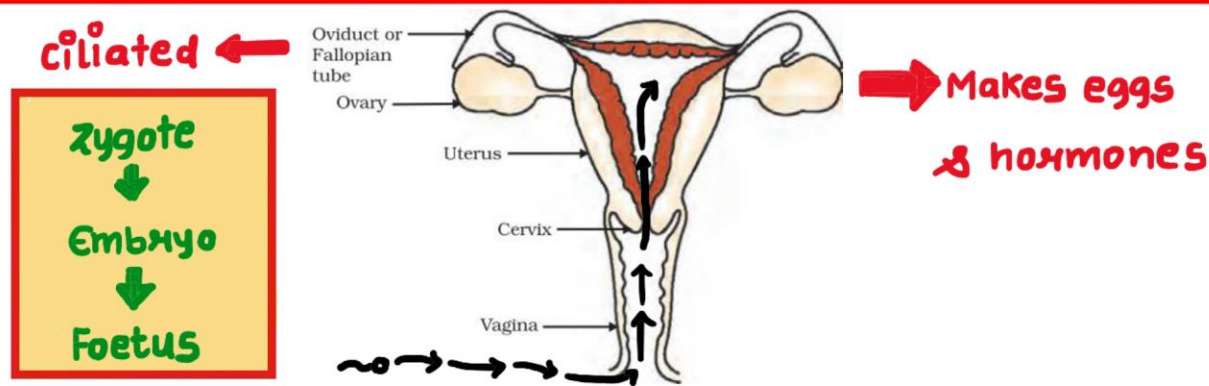
- Mental preparedness is also important
- Bacterial STDs: Gonorrhoea, Syphilis
↳ Neisseria ↳ Treponema

Viral STDs: Warts (HPV), AIDS (HIV)

- CONTRACEPTIVE: Physical barriers: condoms; oral pills; uterine: copper-T; surgical: vasectomy & tubectomy

NOTE: Equal opportunity to both ♂ & ♀ child to be given

FEMALE REPRODUCTIVE SYSTEM

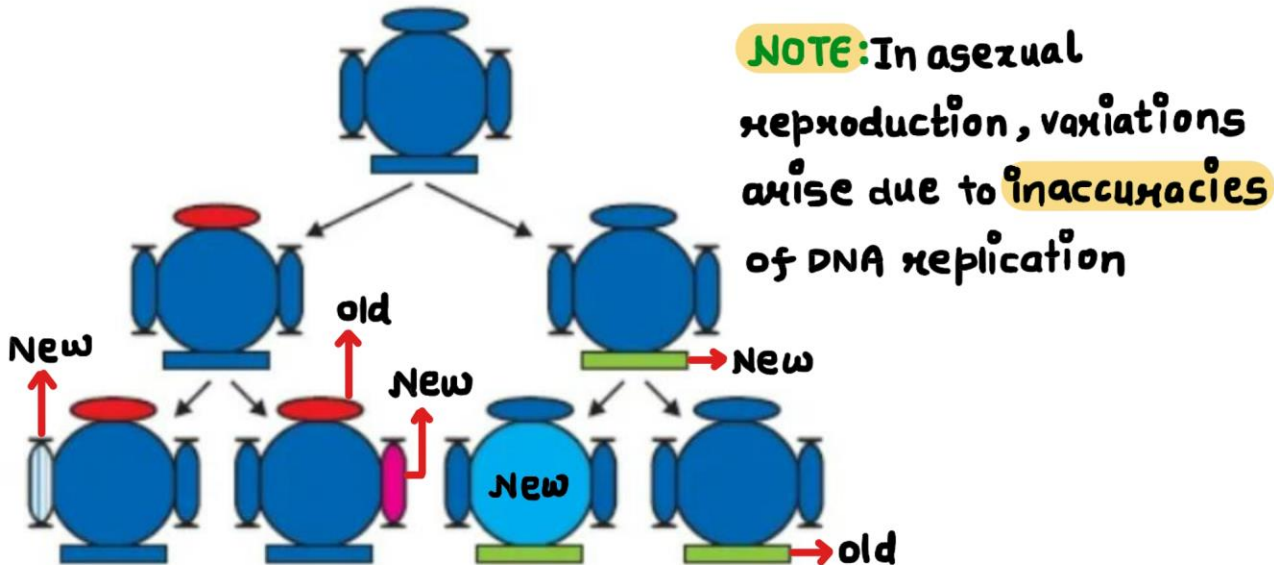


HEREDITY

- Reproduction give rise to organisms which are similar but not identical
- Sexual reproduction leads to more variations

ACCUMULATION OF VARIATIONS DURING REPRODUCTION

- Previous generation inherits (pass on) information of basic body design & some variations to new generation
- New generation will get existing variations from previous generation plus some new/unique variation



NOTE: Such variations gives advantages like heat resistance in bacteria ; these variations are needed for **EVOLUTION**

HEREDITY

- Process by which characters/traits are passed on from parents to offsprings



Free



Attached

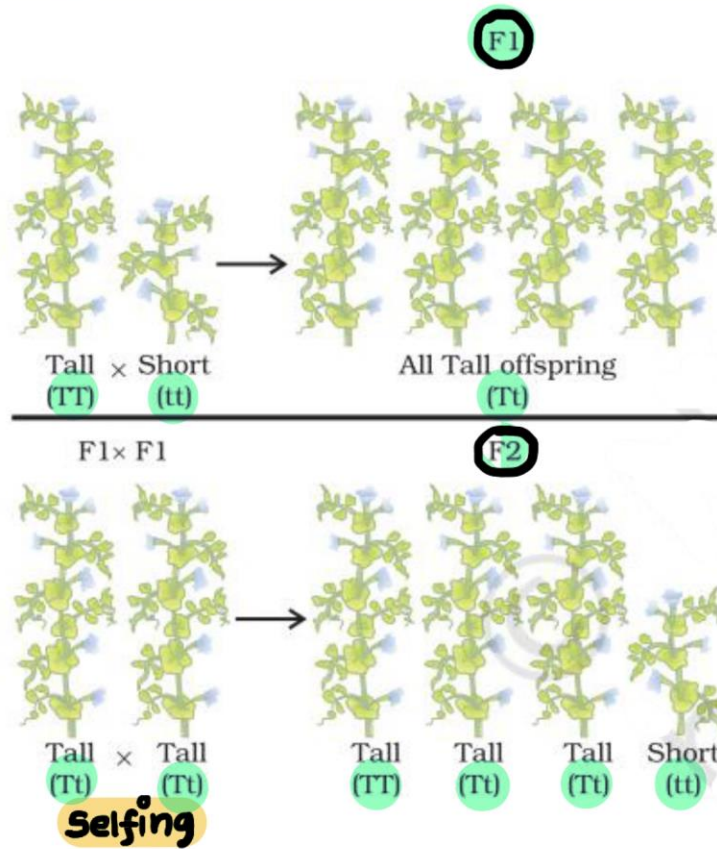
Hair colour: Black, brown

Earlobes: Free, attached

- Mendel (1822-1884), studied the inheritance of traits in pea
- He used no. of contrasting traits for his experiments

- 1 Round & wrinkled seeds
- 2 Tall & dwarf plant
- 3 violet & white flowers etc.

MONOHYBRID CROSS



- Only one character taken in account i.e., height

TT	Tall
Tt	Tall
tt	Dwarf

F₁: All Tall

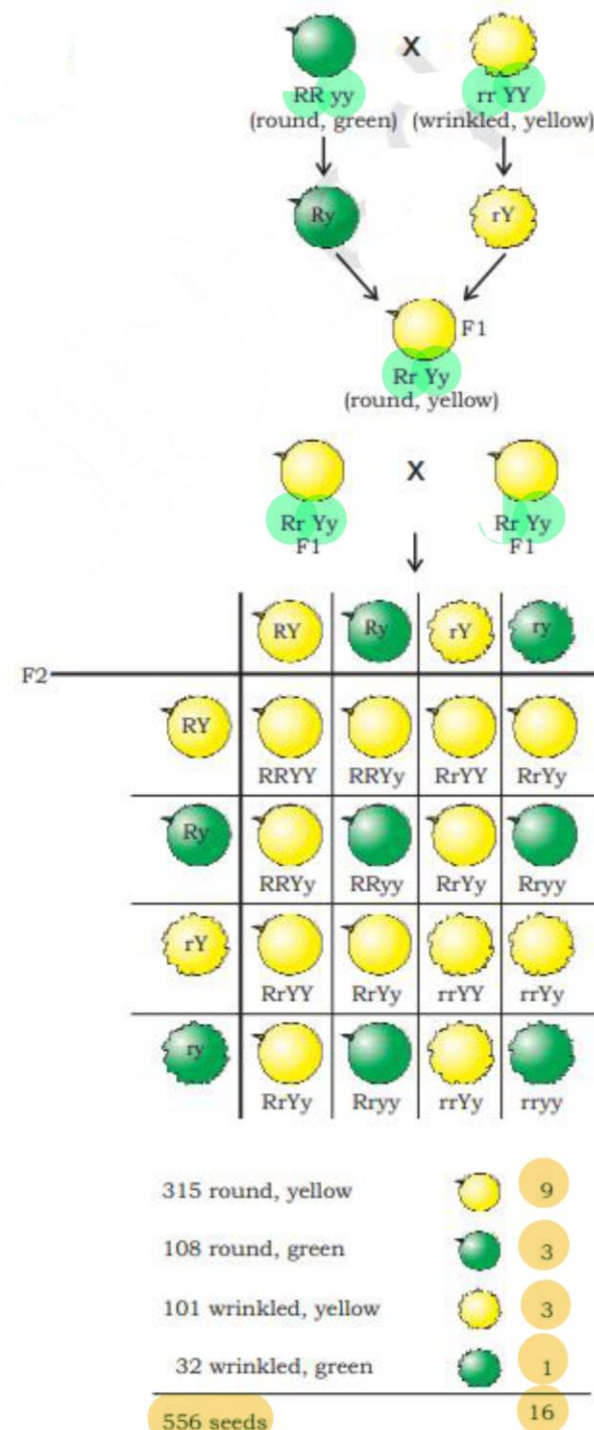
F₂: Tall : Dwarf = 3 : 1

TT : Tt : tt = 1 : 2 : 1

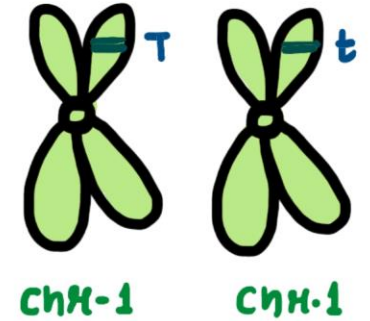
DIHYBRID CROSS

- Two characters studied together i.e., seed colour & shape
- ↓
Round & wrinkled

Yellow & green



- DNA has information to make proteins
- A piece of DNA that can make a protein is called gene



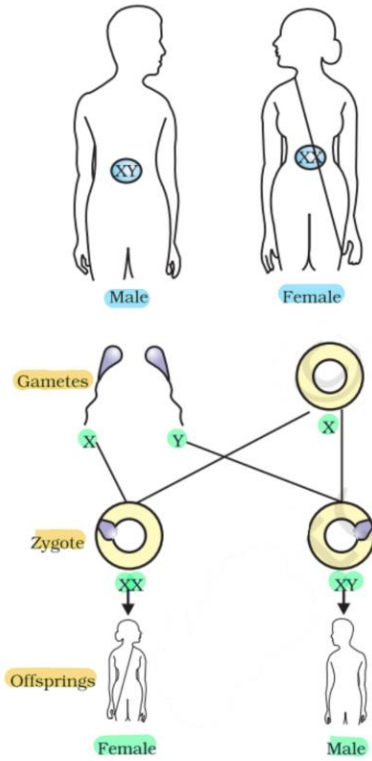
- T & t are alleles i.e., contrasting pair of a gene
- When gamete is formed, it gets only one gene out of these two
- ∴ Each germ cell will have one set of chromosomes

SEX DETERMINATION

- In a few reptiles: Temperature at which fertilised eggs are kept decides sex
- Snails: change sex according to environmental factors
- Humans: sex is determined genetically
- Humans have 23 pairs of chromosomes

22 pairs autosomes
(common in ♂ & ♀)

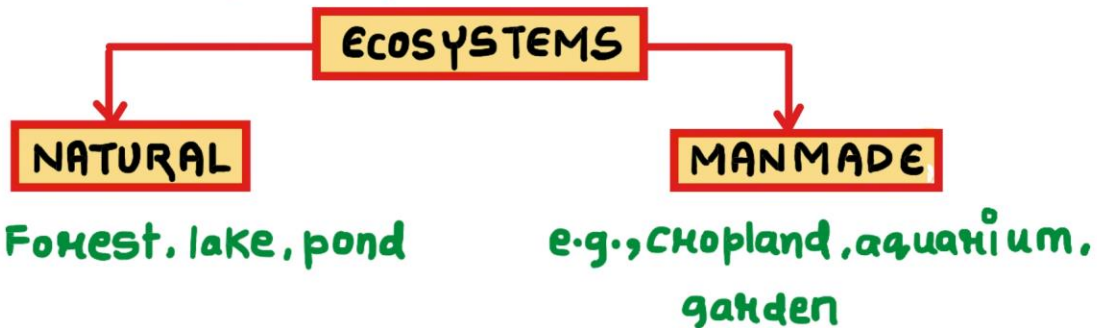
1 pair sex chromosomes
(XX in ♀; XY in ♂)



OUR ENVIRONMENT

ECOSYSTEM & ITS COMPONENTS

- **Environment**: everything that surrounds us
- **Ecosystem**: Biotic + Abiotic components
 - ↓
 - ↳ soil, H₂O, light, Temperature
 - Plants, animals, decomposers

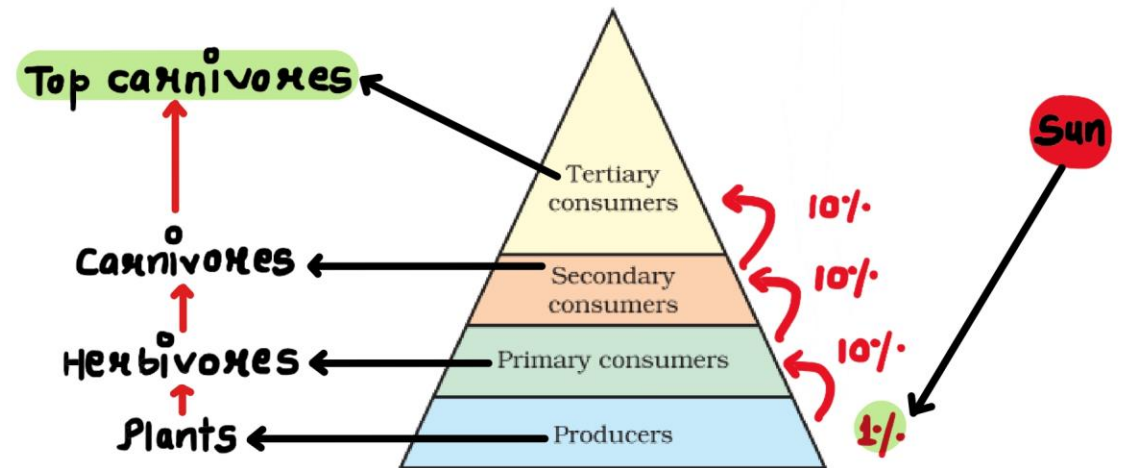


NOTE: Biotic organisms depends upon abiotic factors

- **Producers**: all plants & certain bacteria
- **consumers**: depends directly or indirectly on plants i.e., they can be herbivore, carnivore, omnivore & parasites
- **Decomposers**: mostly bacteria & fungi that can convert organic matter into inorganic matter

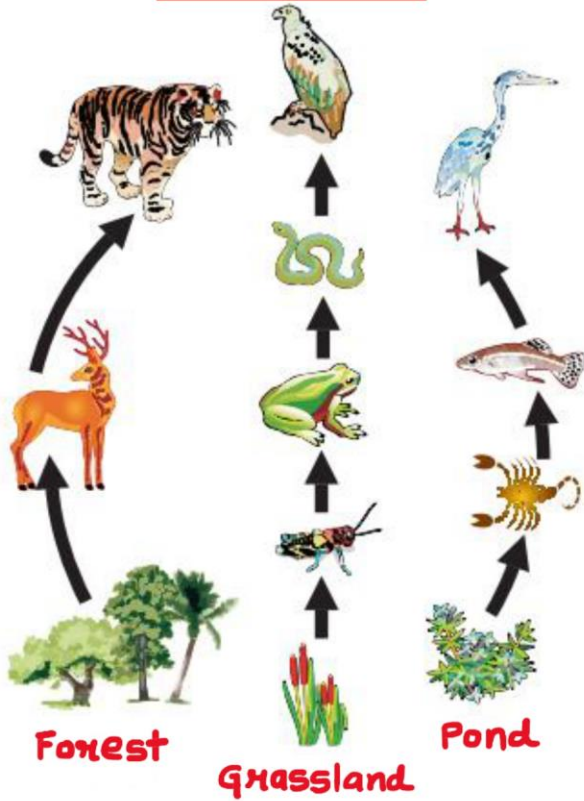
FOOD CHAINS AND WEBS

- Each step of food chain or web represents a trophic level e.g., producers at first trophic level
- Energy flows from one trophic level to another (↓)

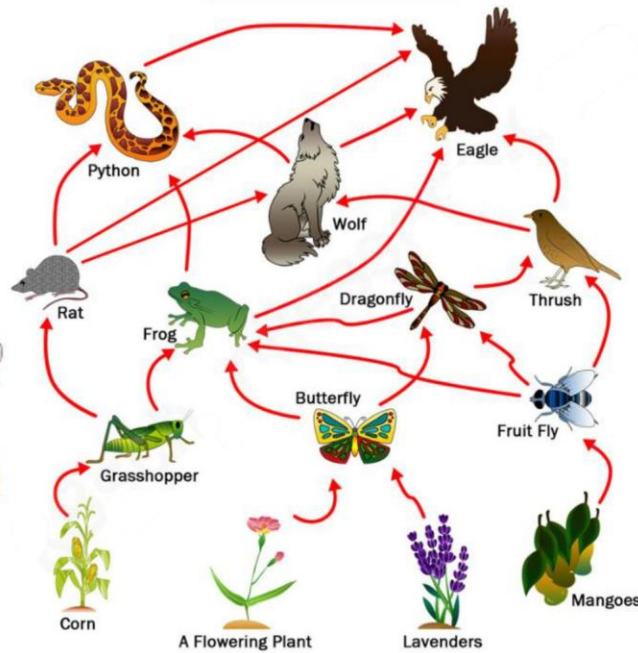


- Energy flows unidirectionally by Lindmann's 10% rule
- Non-biodegradable pesticides & insecticides shows biomagnification in ecosystem
- Limited trophic levels are there in a food chain
- Many food chains link to form food webs ∴ organisms feed on multiple resources

Food chains



Food Web



- O_3 level began to drop in 1980s due to heavy emission of CFC (Chlorofluorocarbon) coming from refrigerators & fire extinguishers
- 1987: UNEP i.e., United Nations Env. Programme made it mandatory to make CFC free appliance

MANAGING THE GARBAGE WE PRODUCE

- Non-biodegradable things like plastic are not degraded by decomposers or saprophytes $\therefore$ they pile up
- $\therefore$ We need to add more & more biodegradable things in our life style e.g., kulhads, paper cups, glass cups etc.

OZONE LAYER AND ITS DEPLETION

- O_2 is beneficial for all aerobic organisms but O_3 is poison; but O_3 shield us from harmful UV rays which causes cancer

